

Recommendations for the Future of Artificial Intelligence: Opportunities, Ethics, and Challenges

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Abstract: Artificial intelligence (AI) is revolutionizing the healthcare industry, allowing for automated decision-making, accurate diagnosis and interpretation in broad disease categories. In this chapter, the opportunities, ethical implications and technical challenges that will shape the future of AI in medicine are explored, focusing on the use of Machine Learning (ML) and Explainable Artificial Intelligence (XAI) for chronic liver disease. Liver cirrhosis is taken as a representative clinical case study: cirrhosis is a chronic liver disease affecting more than 58.4 million people in the world and responsible for 1.4 million deaths in 2021 alone, a rate which has increased by 58.21% compared to 1990. Early-stage cirrhosis is mostly symptom-free, and diagnosis is a key unmet need as AI has a great potential to be used to help the diagnosis. Early diagnosis of cirrhosis is a significant unsatisfied need, as the burden of early cirrhosis is enormous, but it is very likely that the majority of the cases are asymptomatic, and AI has a great potential to be used to help the diagnosis of early cirrhosis. A systematic overview of the ML and XAI methods for Liver Cirrhosis diagnosis is provided with the synthesis of evidence from 35 peer-reviewed studies published in the last 11 years (2015-2025) found after a search of five databases guided by the PRISMA 2020 guidelines. Gradient boosting methods like XGBoost perform best on the structured clinical data with the highest accuracy (85-96%), while SHAP and LIME have shown a consistent ability to highlight platelet count, albumin and age as clinically concordant markers. We critically review longstanding issues, such as dataset uniformity, the lack of external validation of XAI, regulatory science, and clinical translation, and propose evidence-informed recommendations ranging from federated learning, causal AI to regulatory science and human-centred design. The chapter ends with a forward-looking outlook on the ethical and safety requirements that must be met to realize the transformative power of AI on clinical medicine and its governance structures.

Keywords: Artificial Intelligence (AI), Machine Learning (ML), Explainable AI (XAI), XGBoost, SHAP, LIME, Liver Cirrhosis, Ethics, Clinical AI, PRISMA, Federated Learning, Healthcare Recommendations.

INTRODUCTION

AI is the leading edge of the most impactful shift to date in healthcare. There is a convergence of large-scale clinical data, computational power and algorithmic innovation, leading to systems that perform diagnostics, prognostics, drug discovery and treatment optimization as good or better than humans can do, if they are experts in these fields [10,11]. However, the promise of AI in healthcare comes with significant opportunities, a significant ethical quandary and immense technical challenges that will

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need to be carefully considered should AI systems be successfully and responsibly applied to the clinical sphere on a large scale [15,52].

The next three are the dimensions of the future of clinical AI. The opportunity dimension: AI systems can process large amounts and types of medical information that are far beyond the cognitive ability of humans, uncover subtle patterns in multiparametric datasets, and provide consistent and scalable diagnostic assistance across healthcare systems, regardless of their level of resources [10,14]. Second, an ethical dimension: AI systems in healthcare make consequential decisions which impact patient lives, and their use brings with it important questions around algorithmic transparency, fairness, accountability, and patient and clinician rights to understand and challenge the automated decision [52,54,69]. Third, the challenge dimension: from lab bench to clinical implementation, the challenge dimensions of moving from lab bench to clinical implementation are not yet met; there is a need for rigorous validation, regulatory clearance, EHR integration and continued interdisciplinary cooperation that most existing AI research has not achieved [11,45,46].

In this chapter, these three dimensions are discussed with respect to a single concrete clinical case study intelligent diagnosis of liver cirrhosis using ML and XAI and within the specific context of health promotion and disease diagnosis. These three dimensions are discussed in the context of health promotion and disease diagnosis with respect to the single concrete clinical case study intelligent diagnosis of liver cirrhosis using ML and XAI. Liver cirrhosis is one of the most widespread and deadly chronic diseases worldwide, with more than 58.4 million people diagnosed with the condition, and 1.4 million deaths recorded each year [7]. On the other hand, as one of the most diagnostically challenging diseases in medicine, the early signs of the disease are silent, the gold standard diagnostic test, liver biopsy, is invasive, risky and not applicable at scale [8]. It's a perfect space to see the power of clinical AI transforming and how it's implemented in the real world, responsibly. The liver carries out more than 500 physiological roles such as the removal of toxins, production of bile, storing glycogen and producing coagulation factors [1]. Repeated exposure to hepatotoxic substances, mainly ethanol, hepatitis B and C viruses and lipid overaccumulation, causes repeated hepatocyte injury, repair and fibrogenesis [2]. This process of the loss of normal parenchyma and the substitution of fibrous scar tissue results in cirrhosis [3]. Patients can have no symptoms of the disease even when there is a good amount of fibrosis during the compensated stage [4]. Hepatic reserve is critically reduced at the time of the onset of decompensating symptoms (ascites, hepatic encephalopathy or variceal hemorrhage) and median survival without transplantation is less than 2 years [5]. AI has the potential to make significant contributions in this diagnostic gap.

Machine learning algorithms have shown to be more accurate in discovering complex non-linear relationships in high dimensional clinical data [10,72]. But powerful models of machine learning are inherently opaque, which presents a critical ethical obstacle: clinicians must not be able to take a probability decision based on a diagnostic prediction without knowing what it is based on [12,17]. The

solution is provided by explainable AI systems such as SHAP [19] and LIME [20] that can easily explain and audit the model's decisions in a clinical setting.

CHAPTER ORGANISATION

Chapter organisation shows all sections in a serial view, that describes a detailed analysis of the applications of AI in clinical medicine. The chapter uses the case study of Liver Cirrhosis. Section 1 of the chapter first frames the broad opportunities, ethics and challenges related to using AI in clinical medicine. A systematic review methodology is described that forms the basis of evidence collected in section 2. The clinical context of liver cirrhosis is given in section 3 and it is used to rationalize and explain the use of AI for diagnosis. In Section 4 we introduce the technical underpinnings of Machine Learning and in Section 5 we cover the ethical considerations of Explainable AI. Section 6 compares the performance of the algorithms across studies reviewed and in Section 7, offers in-depth methodological analysis. Section 8 compiles the findings of the evidence-based future directions for AI in healthcare and Section 9 presents the summary and future directions. This organizational flow is shown in figure 1.

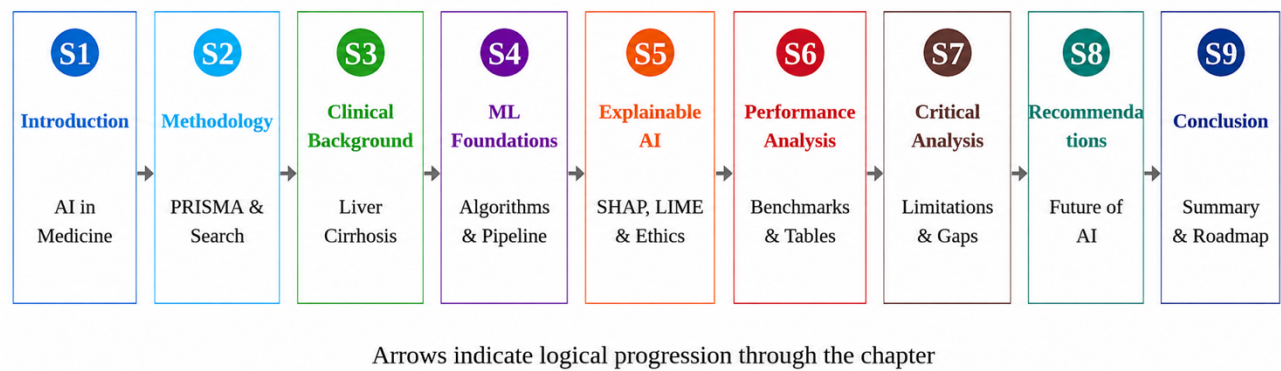


Fig. (1) Organisational flow of this chapter, illustrating the nine sequential sections and their logical progression from clinical problem framing through systematic evidence synthesis to evidence-based recommendations for the future of AI in medicine.

SYSTEMATIC REVIEW METHODOLOGY

Review Protocol and PICO Framework

This chapter uses guidelines for Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA 2020). The clinical question was formulated as follows: in patients with suspected or confirmed liver cirrhosis, how accurate and useful are ML and XAI-based models for diagnosis, staging, or prognosis compared to conventional clinical scoring systems (Child-Pugh, MELD, FIB-4) and unimodal ML models? The methodological quality was evaluated by the PROBAST+AI tool [45] that evaluates the risk of bias in four domains: participants, predictors, outcome, and analysis.

Literature Search Strategy

An extensive electronic search has been performed on PubMed/MEDLINE, IEEE Xplore, Scopus, Web of Science and Google Scholar databases from January 2015 to December 2025. The following search string was used to perform a Boolean search:

("liver cirrhosis" OR "hepatic fibrosis" OR "liver disease") AND ("machine learning" OR "deep learning" OR "artificial intelligence" OR "ensemble") AND ("diagnosis" OR "classification" OR "staging") AND ("explainable AI" OR "XAI" OR "SHAP" OR "LIME" OR "interpretable")

Manual backward reference tracking of included studies was also conducted.

Inclusion and Exclusion Criteria

Studies were considered for inclusion when they involved: (1) an ML or XAI model for liver cirrhosis diagnosis, staging or prognosis; (2) structured clinical/laboratory data, medical imaging, or eHR; (3) quantitative performance metrics; (4) peer-reviewed journals or major conference proceedings; and (5) English language. Studies that focused only on non-cirrhotic hepatic disease and/or editorials with no original data, without adequate methodological detail or using only traditional statistical methods and without ML components were excluded.

The selection of studies and the PRISMA Flow chart.

As shown in fig. (2) a total of 847 records were found from the electronic search. The abstracts were subsequently screened by title/abstract with 487 excluded, leaving 633 abstracts for further screening. Of the 146 articles that were full text assessed, 111 were excluded (non-cirrhosis focus: 34; insufficient metrics: 28; review articles: 22; non-English language: 15; other: 12). Thirty-five papers were included in the analysis. Small sample size and lack of external validation were the most common causes of high risk of bias associated with PROBAST+AI assessment in 22 out of 35 studies (62.9%).

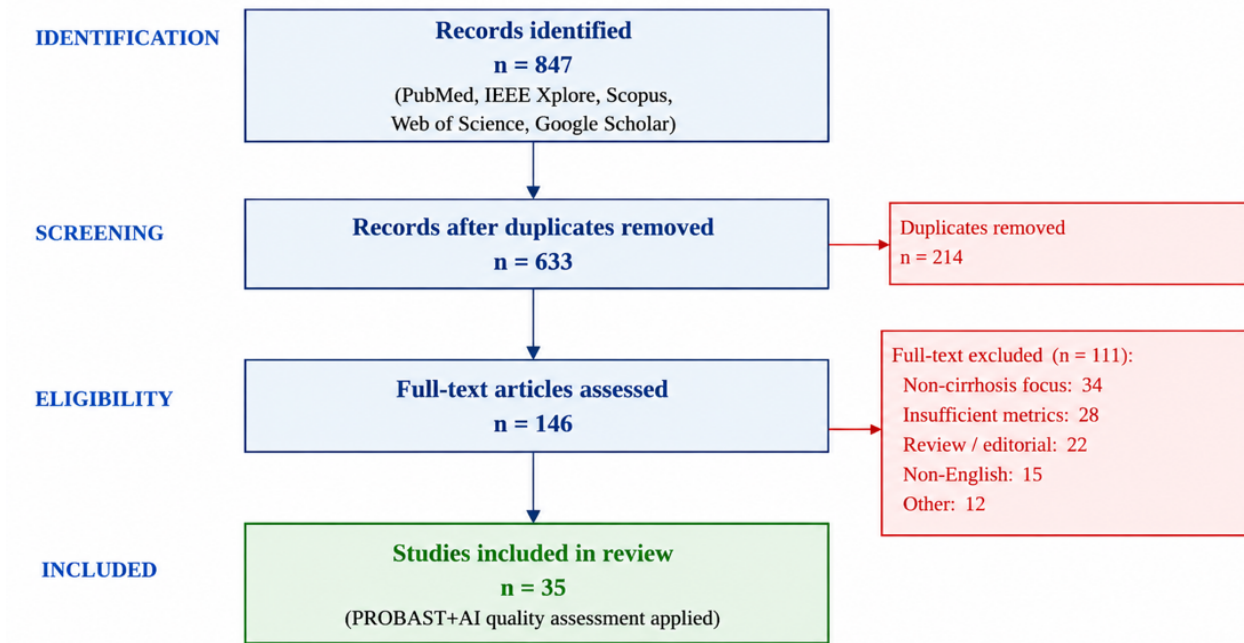


Fig. (2) PRISMA 2020 flow diagram illustrating the study selection process. A total of 35 studies met all inclusion criteria from an initial pool of 847 identified records.

CLINICAL BACKGROUND: LIVER CIRRHOSIS AS A CASE STUDY FOR AI

Pathophysiology and Disease Progression

Chronic liver injury of various nature leads to liver cirrhosis. The cyclic necrosis of the hepatocytes leads to activation of the hepatic stellate cells, synthesis of collagen and other extracellular matrix proteins, which creates fibrous septa that alter the architectural organization of the liver [3]. Cirrhosis is an irreversible stage that is different from the earlier stages (steatosis and fibrosis), because cirrhosis involves the development of regenerative nodules surrounded by fibrous bands, which significantly alters the blood flow and function of the liver. The natural history consists of compensated (silent) and decompensated (symptomatic) phases, with decompensation being a watershed in which mortality becomes very high after 1 year [5]. This is a progressive spectrum, as shown in figure 3.

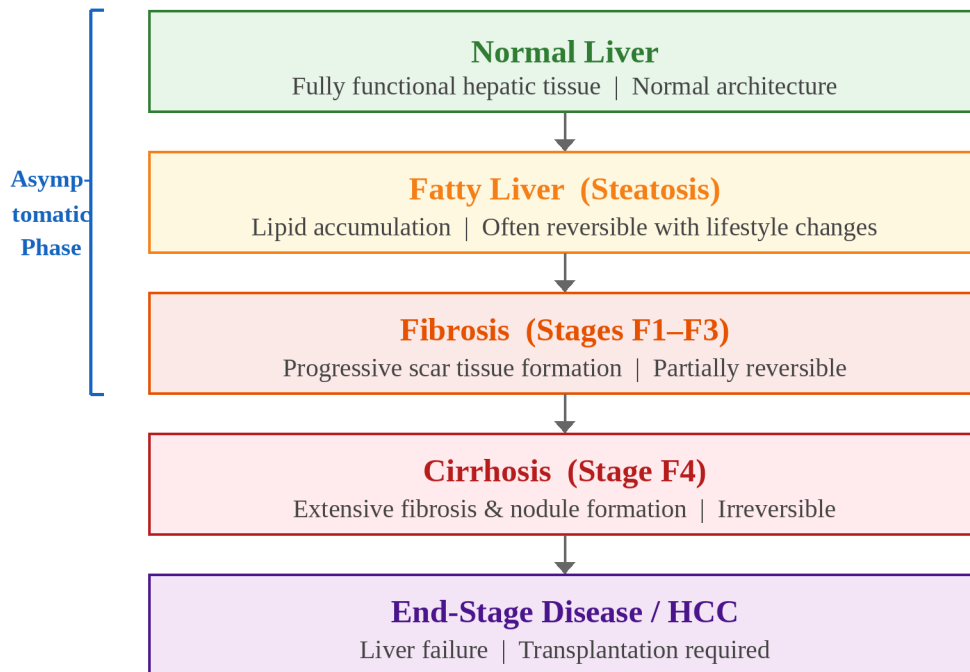


Fig. (3) Progression of liver disease from normal hepatic tissue to end-stage liver failure. Early stages are often asymptomatic, underscoring the critical opportunity for AI-assisted early detection.

Why Liver Cirrhosis is an AI Opportunity.

Liver cirrhosis represents the essence of the AI opportunity in medicine for three reasons. First, the diagnostic gap: The conventional non-invasive tools have an AUROC ranging from 0.78 to 0.86 for advanced fibrosis and are poor at intermediate stages [9] which is a precise need that ML can meet by incorporating multiparametric clinical data. Second is the scale challenge: India sees one million new cases each year [6], and 58.4 million in the world [7] - scalable and reproducible diagnostic methods are required which AI can deliver. Third, the interpretability imperative: cirrhosis diagnosis affects life-altering decisions, such as transplant listing, which means that not only do clinicians need to be able to make predictions, but they also need to be able to explain or justify their choices, thereby affording an ideal space in which to illustrate the value and limitations of XAI [17, 52].

MACHINE LEARNING FOUNDATIONS FOR CLINICAL DIAGNOSIS

Algorithmic Classification

The following grade of complexity of ML algorithms should be used for liver cirrhosis diagnosis:

- 1) **Conventional Statistical Models:** Models with linear relationships between features (logistic regression and linear discriminant analysis) have built-in interpretability without being as affected by complex interactions between features.
- 2) **Tree Based Ensemble Methods:** The robustness and the consistent accuracy (75–89%) have been observed for Random Forest [23]. Structured clinical data can be used to measure accuracy on this group of gradient boosting machines, which include XGBoost [22], LightGBM [31] and CatBoost [32] and are able to achieve 85–96% accuracy.
- 3) **Support Vector Machines:** SVM's are the optimal separating hyperplane in kernel transformed feature space. Accuracy of a binary cirrhosis classification is between 70–85% and not easily scaled to large data sets [27].
- 4) **Deep Learning:** CNNs and attention-based architectures suitable for automated feature extraction from medical images [25,61]. These have large and elaborate annotated data, and do not have self-explanatory properties.
- 5) **Hybrid Stacked Ensembles:** The most recent multimodal ensembles for cirrhosis staging use CNN feature extraction and gradient boosting meta-classifiers [33, 74].

Data Preprocessing Pipeline

- 6) **Missing Data Imputation:** In clinical data sets, between 5 and 30% of the data is missing; thus, missing data imputation is a question that arises and multiple imputation by Chained Equations (MICE) is the methodological gold standard [36].
- 7) **Feature Scaling:** Z score standardization and mini-max normalization is important for distance-based algorithms [37].
- 8) **Class Imbalance Management:** In many datasets, the number of cases with cirrhosis is 5-20%. SMOTE [35] creates synthetic minority class instances, significantly increasing the recall.
- 9) **Feature Selection:** Recursive feature elimination, LASSO regularization and impurity measures from trees decrease dimensionality and improve interpretability [38].

Validation Strategies

- 10) **K-Fold Stratified Cross-Validation:** This is the most common validation method used in the reviewed literature [39]. Maintains class proportions through folds.
- 11) **Bootstrapping:** A resampling method with replacement that yields empirical performance distributions for constructing confidence intervals is called bootstrapping [43].
- 12) **External Validation:** External validation, the most critical methodological step to achieve clinical translation was accomplished in only 4 of 35 (11.4%) reviewed studies (45, 46).

EXPLAINABLE AI: ETHICS, INTERPRETABILITY, AND CLINICAL TRUST

Being interpretable is an ethical requirement.

Ethical transparency is an unwritten rule when it comes to medical decisions and using AI. When it comes to medical decisions and using AI, ethical transparency is an unwritten rule. Clinicians, patients, and regulators are entitled to know the grounds for the decisions made or influenced by AI systems when they impact on patient health, meaning it is essential for them to be informed of this [52, 54, 69]. Consequential automated decisions are covered by the EU GDPR Article 22, which requires a right to explanation. Algorithms must be disclosed in clinical tools as per the FDA's AI/ML Software as a Medical Device Action Plan [46, 69]. The American Medical Association supports AI supports providing clinician accessible, auditable explanations [13].

Several key principles identified in the AI ethics literature that should guide the deployment of clinical AI have emerged: beneficence (AI can have a beneficial effect on the patient); non-maleficence (AI must not lead to harm, either due to incorrect or poor predictions, or to bias); justice (AI benefits must be equitably distributed); and autonomy (the decision to use AI must be retained with the clinician and patient) [52, 54, 70]. The technical means for achieving these ethical principles is called interpretability without which beneficence can't be confirmed, harm can't be recognized, bias can't be identified, and autonomy can't be sustained [15, 55, 68].

Ghassemi et al. [17] have critically looked at the 'false hope' of existing XAI approaches, as plausible post hoc explanations can lead to clinician overconfidence in incorrect predictions, a patient safety hazard which highlights that interpretability does not equal clinical validity. For high-stakes decisions, Rudin [68] maintains that the interpretation of a model should be prioritized over explanations of a black-box model that were built afterwards. These views highlight not only the technological aspects of XAI but also its ethical implications and potential impact on safety.

SHAP: Shapley Additive Explanations

In [19] a mathematically sound attribution framework based on cooperative game theory is provided, called SHAP. A prediction coalition game is played for each feature; the Shapley value is the average of all marginal contributions to the coalition of all the features. The SHAP value for instance x and feature i is the following:

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|! (|N| - |S| - 1)!}{|N|!} [f_{S \cup \{i\}}(x) - f_S(x)] \quad (1)$$

SHAP possesses three properties that make it useful: it is efficient (the sum of all the values equals the prediction minus the baseline), it is symmetrical (if two features make an equal contribution, they get the same value), and the null player property (if the contribution of a feature to the prediction equals zero, then its contribution to the prediction equals zero) [15]. Platelet count, albumin and age are consistently and strongly correlated with liver cirrhosis in SHAP studies, as per the clinical indices (FIB-4, APRI) already established in liver cirrhosis studies [21, 34].

LIME: Local Interpretable Model-Agnostic Explanations

LIME [20] creates local linear models of the behaviour of complex models near a particular prediction instance. The procedure comprises of four stages: (1) creating perturbed samples around the instance; (2) acquiring black-box model predictions; (3) using a weight based on proximity to assign weights to the samples; (4) fitting an interpretable surrogate model. LIME is model-agnostic, and can be used on tabular, image, and text data. The main drawback of its ethics is that they are unstable, that is, the results of the explanation depend on the values of the perturbation parameters, and that the level of fidelity in the local explanation does not imply a corresponding level of fidelity in the global explanation [17, 57].

The Importance of Counterfactual and Causal Explanations

The goal of counterfactual explanations [66] is to find the smallest change to the input features that would lead to a different prediction, which can then be used to inform patient-specific clinical decisions (such as the amount of bilirubin that needs to be reduced to change a high-risk patient's classification). Ethically, counterfactuals shift from a diagnostic function to a clinical intelligence that helps to support patients' autonomy and treatment decisions. XAI is another example of the national-level recognition of the importance of interpretability for responsible AI in critical domains, this time from DARPA [67].

PERFORMANCE ANALYSIS AND COMPARATIVE BENCHMARKING

Performance Metrics Framework

The following measures are mentioned in Table 1 for assessing the accuracy of a liver cirrhosis prediction model: discrimination, calibration and clinical utility. While discrimination assesses the ability of the prediction model to distinguish between people who have liver cirrhosis and those without liver cirrhosis, calibration assesses how reliable its probability predictions are whereas clinical utility assesses its applicability in real-world medical practice.

Table 1 Performance metrics for liver cirrhosis diagnosis models

Category	Metric	Formula	Clinical Interpretation	Ref.
Discrimination	Accuracy	$(TP+TN)/(TP+TN+FP+FN)$	Overall correctness; misleading for imbalanced classes	[40]
	Sensitivity	$TP/(TP+FN)$	Identifies true cirrhosis cases; minimizes false negatives	[40]
	Specificity	$TN/(TN+FP)$	Correctly excludes non-cirrhosis; minimizes false positives	[40]
	F1-Score	$2 \times (Prec \times Rec) / (Prec + Rec)$	Harmonic mean; preferred for imbalanced datasets	[40]
	AUC-ROC	Area under ROC curve	Discrimination across all thresholds (0.5=random; 1.0=perfect)	[41]
	AUC-PR	Area under PR curve	Superior to AUC-ROC for highly imbalanced classes	[42]
Calibration	Brier Score	$(1/N) \sum (p_i - \hat{p}_i)^2$	Probability accuracy; lower is better	[43]
	Calibration Slope	Observed vs. predicted slope	Ideal = 1.0; <1 indicates overconfident predictions	[43]
Clinical Utility	Net Benefit	$TP/N - FP/N \times pt/(1-pt)$	Decision curve analysis at threshold pt	[44]

Algorithm Performance Comparison

The conventional statistical models achieve the accuracy range of 65-78% across the 35 reviewed studies; SVMs 70-85%, Random Forest 75-89%, Gradient Boosting Ensembles 85-96% and Deep Learning 87-97% in imaging applications. A cross-study comparison is given in Table 2 and graphical representation is given in fig. (4).

Table 2 Comprehensive comparison of ML algorithms for cirrhosis diagnosis

Year	Algorithm	Dataset (n)	Acc. (%)	Sens. (%)	Spec. (%)	AUC-ROC	XAI Method	Ref.
2022	Random Forest	Private (418)	78.2	75.4	80.1	0.82	None	[26]
2022	SVM	UCI (583)	71.4	95.2	97.5	0.97	None	[27]
2022	Logit Boost	ILPD (583)	89.3	87.6	90.2	0.91	None	[28]
2020	Ensemble	VA (12,383)	83.7	81.9	84.5	0.88	None	[29]
2021	Deep Learning	EHR (24,156)	87.4	85.2	88.3	0.90	Attention	[33]
2020	CART	Egyptian (4,423)	95.5	94.8	95.9	0.96	None	[30]
2024	XGBoost	Kaggle (418)	91.19	94.0	100†	0.98	SHAP, LIME	[34]
2021	LightGBM	Korean (2,148)	90.2	89.4	90.8	0.94	SHAP	[31]
2023	CatBoost	Chinese (1,876)	89.8	88.2	90.9	0.93	LIME	[32]
2023	XGBoost+SHAP	PBC (312)	88.7	86.5	90.2	0.92	SHAP+Anchors	[21]
2025	XGBoost+SHAP	Multi-site (2,890)	93.1	92.4	93.8	0.97	SHAP+LIME	[76]

† 100% specificity on single-etiology, female-skewed dataset (n=418, PBC trial). No external validation. Generalisation to diverse populations unconfirmed.

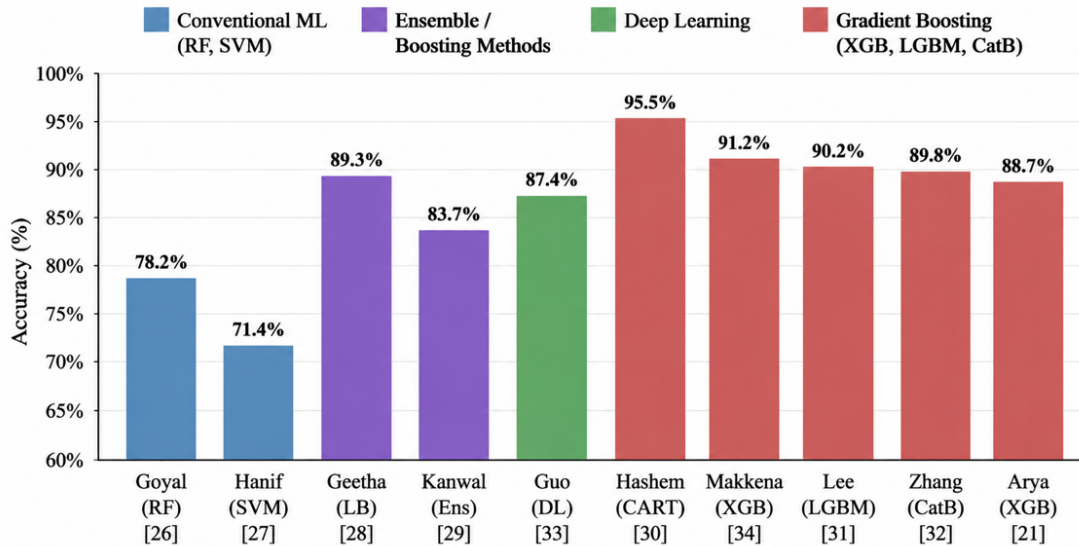


Fig. (4) Accuracy comparison across ML algorithm classes in reviewed studies. Gradient boosting methods consistently outperform classical approaches on structured clinical data.

CRITICAL ANALYSIS AND METHODOLOGICAL LIMITATIONS

Dataset and Demographic Bias

The biggest constraint is the use of minor, single centre, demographically homogenous data sets. Despite being unrepresentative of the majority viral hepatitis B and C cirrhosis populations, the Kaggle cirrhosis dataset (n=418, 89.45% female, from a trial of PBC) has been used in multiple benchmark studies [7, 58]. This geographic/aetiological bias is that high-performing models might not work well in the populations with the highest disease burden. The risks of systematic overfitting are introduced by size constraints of datasets: internally validated estimates of high-capacity algorithms trained on a smaller number of patients (typically 300) often do not replicate externally [39]. There is lack of reporting of calibration (nearly everywhere) and no way to assess the reliability of predicted probabilities for clinical risk stratification [43].

Validation and Statistical Limitations

Among the 35 studies reviewed, only 4 (11.4%) studies did external validation; none of which studies were prospective cohorts, the most important methodological limitation for clinical translation [45, 46]. Reported performance figures should be considered provisional until prospective, multi-site validation is done according to the TRIPOD+AI and PROBAST+AI criteria. Class imbalance is systematically mismanaged: Most studies are not corrected by SMOTE and report accuracy [35] without SMOTE, which leads to misleading high accuracy scores of models predicting the majority class. Less than 40% of studies report systematic hyperparameter optimization, which means that unquantified confounding enters the picture.

XAI Reliability and the Ethics of Explanation

Explainability does not mean reliability, clinical validity. SHAP explanations are shown to have shifted under the influence of noise in the data: small noise changes in the data can lead to significant changes in the attribution vectors, making them unreliable as consistent decision support tools [19]. This is the 'explainability trap', where plausible, but wrong explanations lead to overconfidence in wrong predictions, as has been shown in experiments [17, 66]. These risks are not just technical; they are ethical as patients can be harmed by using and implementing unreliable XAI tools in clinical settings. Only 6 out of 35 reviewed studies conducted expert agreement assessments of outputs of XAI [21].

Clinical Translation and Governance Barriers

The algorithm development to clinical use continuum includes several stages that are not fully explored in the current literature. Just less than 10% of the reviewed studies consider EHR interoperability (HL7 FHIR standards) [46]. There remains a regulatory gap: There is a need for continued post-market model drift monitoring with the FDA SaMD framework, however, no study reviewed had included a monitoring strategy. Ethical issues like algorithmic bias, privacy and protection of data under GDPR and HIPAA and liability when it comes to errors made during diagnosis remain consistently overlooked [52, 54, 69].

RECOMMENDATIONS FOR THE FUTURE OF AI: OPPORTUNITIES, ETHICS, AND CHALLENGES

The critical analysis of liver cirrhosis AI in a case study can be expressed in concrete and evidence-based recommendations for the three items mentioned in the chapter title: what must be done, what must be guarded and what must be overcome.

Opportunities: Technical Advances

- 13) **Data Fusion of Multimodal Data:** Future AI systems must integrate the T2-weighted MRI texture features, serum fibrosis panels, clinical scoring systems and histopathological data within the same architectures. For transformer-based models, attention across modalities [64] has been suggested as a possible solution to the multiparametric integration problem [25, 57] that arises.
- 14) **Federated Learning for Scale and Privacy:** Federated learning [48, 60, 73] is a way to train models on multiple data silos without the need to centralise data to obtain large, varied cohorts that meet GDPR & HIPAA regulations. This directly tackles the limitation of homogeneity that reduces the applicability of existing cirrhosis AI models.
- 15) **Causal AI and Counterfactual Reasoning:** Moving from correlation to causation with the help of a directed acyclic graph and counterfactual reasoning will help explain modifiable causal factors such as alcohol intake and adhering to antiviral treatment and make the role of AI more therapeutic than passive diagnostic for this, AI can be more than a mere passive diagnostic and become an active therapeutic guide.

- 16) **Transfer and Few-Shot Learning:** If the large general medical datasets are used to transfer knowledge and the models are fine-tuned for each domain, the number of annotations needed is considerably reduced [49, 50, 65]. This is particularly important when rare disease phenotypes or low resource clinical settings where enormous and labelled datasets are not available.
- 17) **Large Language Models in Clinical AI:** New capabilities introduced by the emerging large language models (LLMs) include the ability to synthesize clinical narratives, structure unstructured data from the EHR, and provide natural-language explanations for AI predictions [61, 62]. However, the risk of hallucination and calibration limits must be taken into account before clinical use [62].

Ethical advice for the proper use of AI.

- 18) **Inherent Interpretability Where Stakes Are Highest:** AI systems for high stakes, clinical decisions should be more likely to be inherently interpretable models than black-box models with post hoc explanations [68]. If there is a need for using a black-box model for the performance, explanations must be validated independently for stability and clinical concordance prior to deployment of the model.
- 19) **Fairness and Algorithmic Bias Auditing:** Auditing of AI systems should be done in advance for performance differences between demographic subgroups, such as sex, ethnicity, age, socioeconomic status. The Kaggle cirrhosis dataset is a case of systematic bias that is particularly evident in the dataset due to the ratio of male to female. The Kaggle cirrhosis dataset is a case in point for how much systemic bias can be found in a dataset, in this instance, due to the ratio of males to females.
- 20) **Patient and Clinician Autonomy:** AI recommendations should be used as a decision aid, not a decision maker. Clinical workflows need to continue to have human meaning, patients should be able to challenge AI-driven decisions [52, 54, 69].
- 21) **Data Sovereignty and Privacy:** Blockchain-enabled secure data sharing [51] and federated learning [48] are technical solutions that can be used to protect patient privacy while allowing multi-institutional access to data needed for generalizable AI models.

Challenges: Implementation Roadmap

- 22) **Prospective Multi-Centre Validation:** Multi-Centre prospective trials according to TRIPOD+AI [46] and PROBAST+AI [45] criteria are the most important improvement that is still to be achieved. Even the best models are not ready for clinical use, unless they have been externally validated in the future.
- 23) **Regulatory Science and Governance:** There is a need for close interaction between AI researchers, clinical hepatologists, regulatory agencies (e.g., FDA, EMA, and CDSCO), and patient groups to develop evidence standards and post-market surveillance measures and liability frameworks for clinical AI [46, 52, 69].
- 24) **Human-Centered Design and Workflow Integration:** From the start, Co-design with clinician end-users must include iterative prototyping, contextual inquiry, testing for interoperability with

the EHR and usability evaluation, and must be human-centered design and workflow integration [13, 67]. Technical performance is required, but not enough, for clinical use.

- 25) **Standardized Benchmarking and Reporting:** There is a need for some minimum reporting standards in the field which include the demographic composition of the field, the size of the field, the distribution of the aetiology, the type of validation (internal vs external, prospective vs retrospective), the rates of expert agreement for XAI outputs, and stability metrics for outputs. These standards would allow for some meaningful comparison across studies to be done and speed up evidence synthesis.

Long-Term Vision: Precision Hepatology and Beyond is the Global vision.

The integration of federated ML, causally inspired XAI, the use of wearable biosensor systems, liquid biopsy systems and blockchain systems for data sharing represent a new wave of personalised precision medicine [51, 73, 74]. The suggestions in this chapter are more than technical specifications, they are a moral compass to guide the implementation of AI in healthcare in a manner that is safe, fair, reliable, and ultimately beneficial to every patient. To reach this vision, it is important for ML research, clinical medicine, regulatory science, ethics and health services research to work closely together, driven by the vision of health equity and with sustained efforts over time [7, 52, 75].

CONCLUSION

In this chapter, the opportunities, ethical duties, and technical challenges of the future of AI in medicine have been explored and illustrated with the example of intelligent liver cirrhosis diagnosis. A total of 35 peer-reviewed studies were studied from 2015 to 2025, hence the systematic review, which clearly shows that conventional statistical models (65–78% accuracy), ensemble methods (85–96%) and deep learning architectures (up to 97% accuracy for imaging) achieve different levels of performance. XGBoost interpreted by SHAP and LIME always performs best on structured clinical data as far as both performance and interpretability, and platelet count, albumin, and age have been found to be clinically concordant across independent datasets.

A critical analysis of this reveals that there is still a large discrepancy between the algorithmic benchmarks and clinical use. External validation was only carried out in 11.4% of the reviewed studies. Actually, Dataset demographic homogeneity, class imbalance mismanagement, XAI explanation instability and 'false hope' problem outlined by Ghassemi et al. are real patient safety concerns that need to be addressed before clinical deployment. Calibration reporting is nearly never reported and thus the probability reliability needed for shared decision making is not assessed.

The recommendations in this chapter touch on each of the three elements of the chapter's subtitle. The opportunities range from multimodal data fusion to federated learning, causal AI, transfer learning to integration of large language models. Ethical principles covered are interpretability of algorithms for high-stakes decisions, algorithmic fairness audit, preserving patient autonomy, and protecting data sovereignty. Issues are prospective multi-centre validation, development of regulatory science, design of human-centred workflow, and standardization of benchmarking. They provide an ethical and

clinically sound, as well as technically difficult, forward-looking blueprint for the deployment of AI in health care.

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